

Chapter III

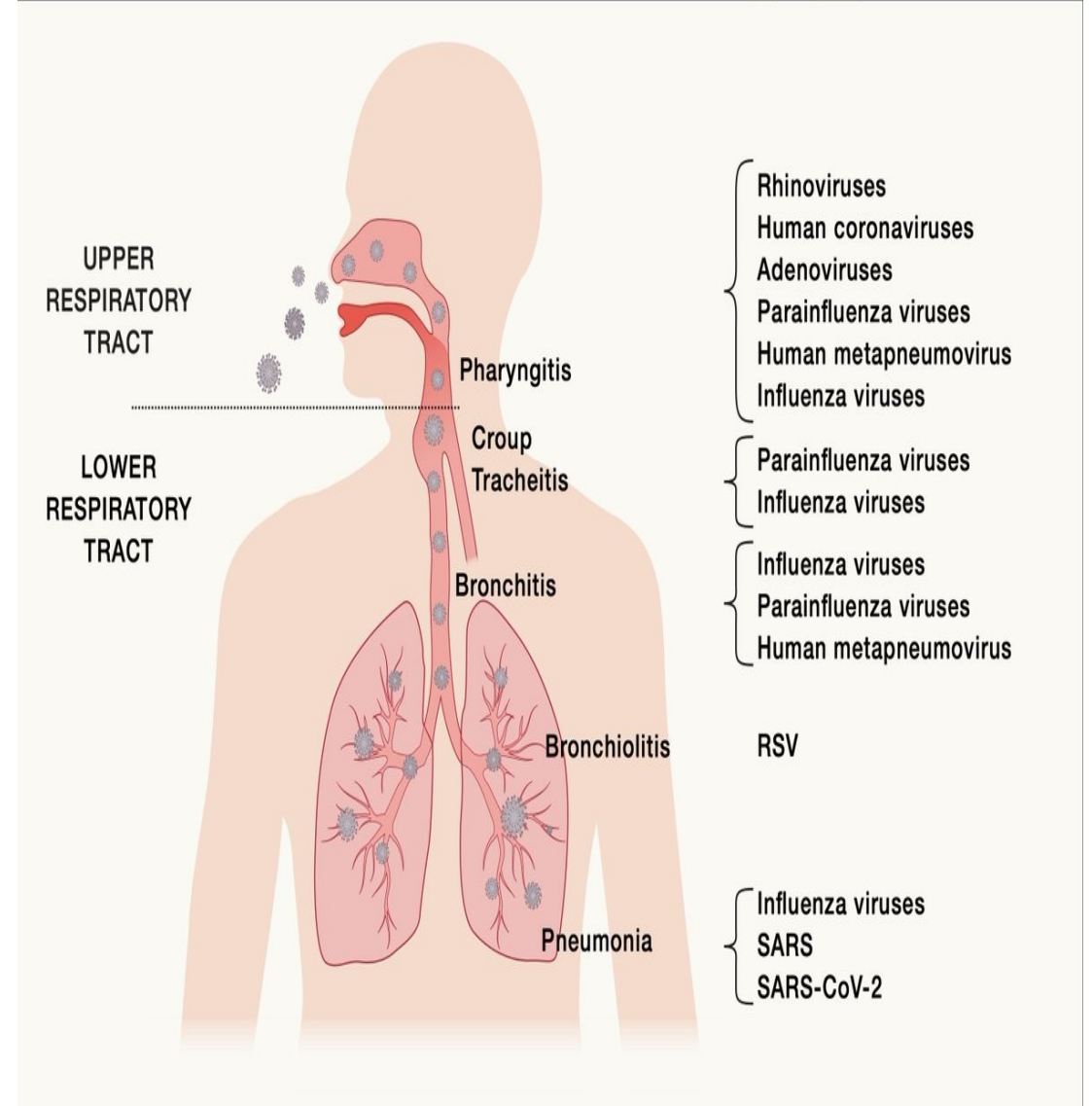
Respiratory Viruses

Objectives

- Explain the properties of viruses causing respiratory diseases
- Describe the pathogenesis of virus causing respiratory diseases
- Explain the clinical features of diseases caused by respiratory viruses
- Describe the diagnosis of viral respiratory infection

VIRAL INFECTIONS OF THE RESPIRATORY TRACT

- The viruses that are major causes of acute respiratory disease (ARD)
 - Influenza viruses,
 - Parainfluenza viruses,
 - Rhinoviruses,
 - Adenoviruses,
 - Respiratory syncytial virus (RSV), and
 - Respiratory coronaviruses.
- Heterogeneous group of viruses shares
 - short incubation period (1–4 days) and
 - a person-to-person mode of spread.

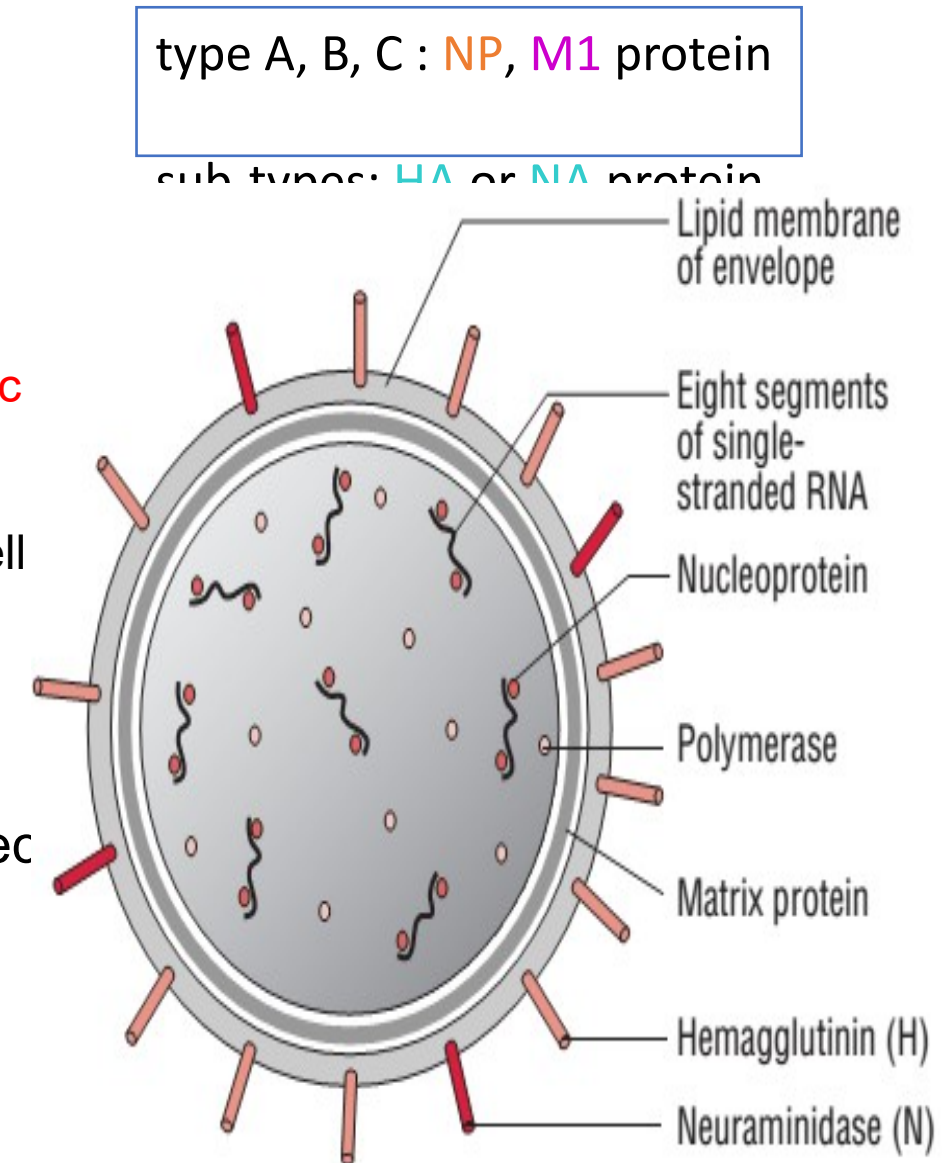


- **Transmission**
 - Direct by infective droplet nuclei, or
 - indirect, by hand transfer of contaminated secretions to nasal or conjunctival epithelium.
- All of these agents are associated with an increased risk of **bacterial superinfection**

INFLUENZA VIRUSES

- Members of the **orthomyxovirus** group, enveloped, pleomorphic viruses
- Classified into **three** major serotypes, A, B, and C, based on different **ribonucleoprotein antigens**.
- **Type A** has greatest virulence and epidemic spread
- Influenza A and B viruses each consist of a nucleocapsid containing **eight segments** of **negative-sense, ssRNA**

- The envelope has Two virus-specified glycoproteins, **hemagglutinin** and **neuraminidase**,
 - virus-specified glycoproteins are antigenic
 - **Hemagglutinin:** agglutinate RBC from certain species (eg, chickens, guinea pigs) in vitro
 - serve as a point of attachment to respiratory cell with **sialic acid**
 - **Neuraminidase:** has role in envelope fusion with host cell membrane and viral release
- Viral isolation in **eggs** or **cell cultures**
- **Hemadsorption** and **hemagglutination inhibition** used to detect presence of virus



Antigenic and genomic properties

- Influenza viruses have two types of antigens
- **Group-specific antigens:**
 - ribonucleoprotein (RNP) antigen, or the “soluble” antigen, or the internal antigen
 - Influenza viruses are divided into types A, B, and C
- **Type-specific antigens:**
 - The surface antigen, or “viral” antigen, or “V antigen” is composed of two virus-encoded proteins,
 - HA and NA, which are the type-specific antigens

Antigenic variations

- Antigenic variation is a unique feature of influenza virus
- HA and NA show variations, RNP antigen and M protein are **stable**
- Antigenic variations are of two types: **antigenic shift** and **antigenic drift**
- **Antigenic shift:** an abrupt, drastic, discontinuous change
 - occurs due to major antigenic changes in HA or NA as a result of **genome reassortment** between different **subtypes**. occurs only in Influenza Virus **A**
 - caused by replacement of the gene for HA by one coding for a completely different amino acid sequence
 - characterized by alteration of virtually all the antigenic sites of the HA
 - Responsible for epidemics as well as pandemics of influenza

❑ **Antigenic Drift:** The gradual, sequential, regular antigenic change

- Occurs due to minor antigenic changes in the HA or NA
- Characterized by changes in **certain epitopes** in the HA
- **New strain** of virus formed,
- Occurs as a result of the accumulation of **point mutations** in the gene.
- Occurs in influenza virus a, b and c and RNA virus escape recognition of host's immune response

TRANSMISSION

- AEROSOL, 10^5 to 10^6 virions per droplet

PATHOGENESIS

- Inhaled influenza viruses reach lower respiratory tract, **tracheobronchial tree** primary site
- attach to **sialic acid** receptors on epithelial cells by **HA** present
- **NA** of the viral envelope may act on the **N-acetyl neuraminic** acid residues in mucus to **produce liquefaction**
- The liquefied mucus help spread the virus through the respiratory tract
- Infection of mucosal cells results in cellular destruction and desquamation of the superficial mucosa.

RESPIRATORY SYNCYTIAL VIRUS

- Classified as a pneumovirus within the **paramyxovirus** family
- Has linear, **non segmented**, (-) ss RNA and Enveloped virus
- Pneumovirus causing **syncytium** formation in cell cultures
- Two glycoproteins mediate attachment and syncytium formation
 - **G** (attachment) and **F** (fusion of the viral envelope) resulting in the formation of large **multinucleated syncytia**
- RSV antigenically stable **one** antigenic serotype with subgroups (A and B) depending on G glycoprotein
 - Subtype A is more virulent than the subtype B
- The leading cause of respiratory tract infection in infants and young children

RSV disease

- RSV primarily infects the **bronchi**, **bronchioles**, and **alveoli** of the lung.
- The illnesses clinically categorized as **croup**, bronchitis, bronchiolitis or pneumonia are extremely common in infants.

Pathogenesis

- Confined to respiratory epithelium
- Enhanced disease in infants may have immunologic basis
- Th2 stimulated cytokines cause injury
- Necrosis and inflammation plugs bronchioles and alveoli

Diagnosis

- Virus isolation, serology and molecular testing

Parainfluenza viruses

- **Enveloped paramyxoviruses** have **neuraminidase** and **hemagglutinin**,
- **Four** serotypes in human
- SS, non-segmented, (-) sense RNA with nucleoproteins P and L and helical symmetry
- Unlike **influenza** virus
 - Replication- cytoplasmic, bud from plasma membrane
 - The antigenic makeup of the four serotypes is relatively stable
 - Significant antigenic shift or drift does not occur.
 - Each serotype is considered separately.

Antigenic and genomic properties

- The virus contains five serotypes: HPIV-1, HPIV-2, HPIV-3, HPIV-4a, and HPIV-4b
- These serotypes show cross-reactions among themselves.
- Types 1,2,3 severe second only to RSV
- Type 4 mild URT
- Type 3 most prevalent- pneumonia infants
- Type 1 and 2 – croup infants (laryngotracheitis)

Pathogenesis PIV

- Infect epithelial cells of URT
- Replication rapid
- Giant cell formation → cell lysis
- Unlike measles and mumps rarely viremia
- Most cases stay in URT → cold like symptoms
- Some LRT ↓
 - Severe 2-3 % cases severe form of laryngotracheobronchitis
 - Death rare
- Laboratory diagnosis by isolation or antigen detection

RHINOVIRUSES

- ♥ Are the common cold viruses containing **ss,(+)RNA** and **non-enveloped**
Picornaviridae
- ♥ At least 150 types are known based on **type-specific antigen** in their capsids
- ♥ Major cause of common cold
- ♥ Sensitive to acidic pH
- ♥ Replicate in vitro at optimum temperature of 33°C
- ♥ Divided into **major** and **minor** groups
- ♥ Most consistently isolated in cultures of human diploid fibroblasts (**M** & **H** strains)
- ♥ The receptor for most rhinoviruses is glycoprotein intercellular adhesion molecule 1 (**ICAM-1**) and **LDLP**(low-density lipoprotein receptor)

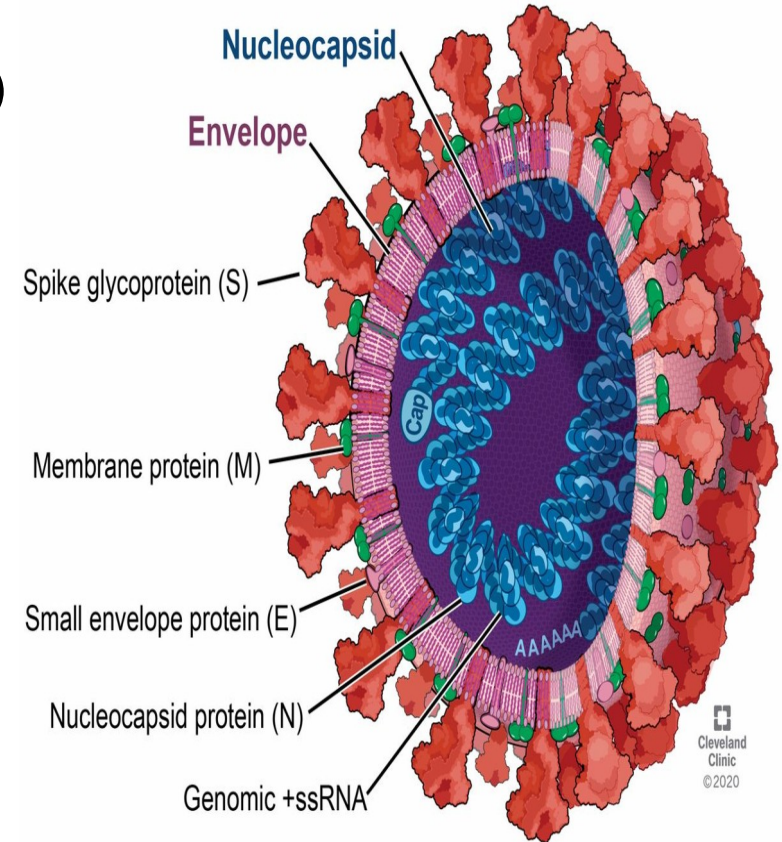
- © Four capsid proteins VP1-VP4 from cleavage of the original protomer VP0
- © One or two nonstructural proteins
- © Surface capsid proteins VP1 and VP3 are major antibody-binding sites.
- © Replication and assembly in cytoplasm
- © Infection: **acute** and **cytolytic** / **persistent**
- © Insensitive to ether, chloroform and nonionic detergents

Treatment and prevention

- ⌈ Multiple serotypes make vaccine different
- ⌈ Pharmaceutical agents block attachment to ICAM (**pleconaril**)

CORONAVIRUSES

- large, enveloped RNA viruses contain unsegmented ss, (+) RNA, has the largest genome among RNA viruses
- cause common colds, LRTI and gastroenteritis in infants
- ***Coronaviridae***
 - ***Coronavirus - Man, animals and birds***
 - ***Torovirus – Ungulates- associated with diarrhea***
- Envelope bearing distinctive, 'club-shaped' peplomers 'crown-like' appearance (Latin, *corona*) 2 gp → make peplomer
 - induce neutralizing and cellular immune responses



Properties of Coronaviruses

The envelope carries three glycoproteins:

- S - Spike protein: receptor binding, cell fusion, major antigen
- E - Envelope protein: small, envelope-associated protein- E1 and E2
- M - Membrane protein: transmembrane - budding & envelope formation in a few types, in some coronaviruses there is an other glycoprotein:
 - HE - Hemagglutinin-esterase
- **Outstanding characteristics:**
 - Cause colds and SARS
 - Display high frequency of recombination
 - Difficult to grow in cell culture

Replication

- cell surface **metalloprotease** and a **sialic acid** are cellular **receptors**
- Coronaviruses are highly species-specific.
- Most of the known animal coronaviruses display a tropism for **epithelial cells** of the **respiratory** or **gastrointestinal** tract.
- High frequency of recombination during replication-unusual for non segmented RNA virus-evolution of new strains
- **SARS** is caused by a novel, new coronavirus

Pathogenesis

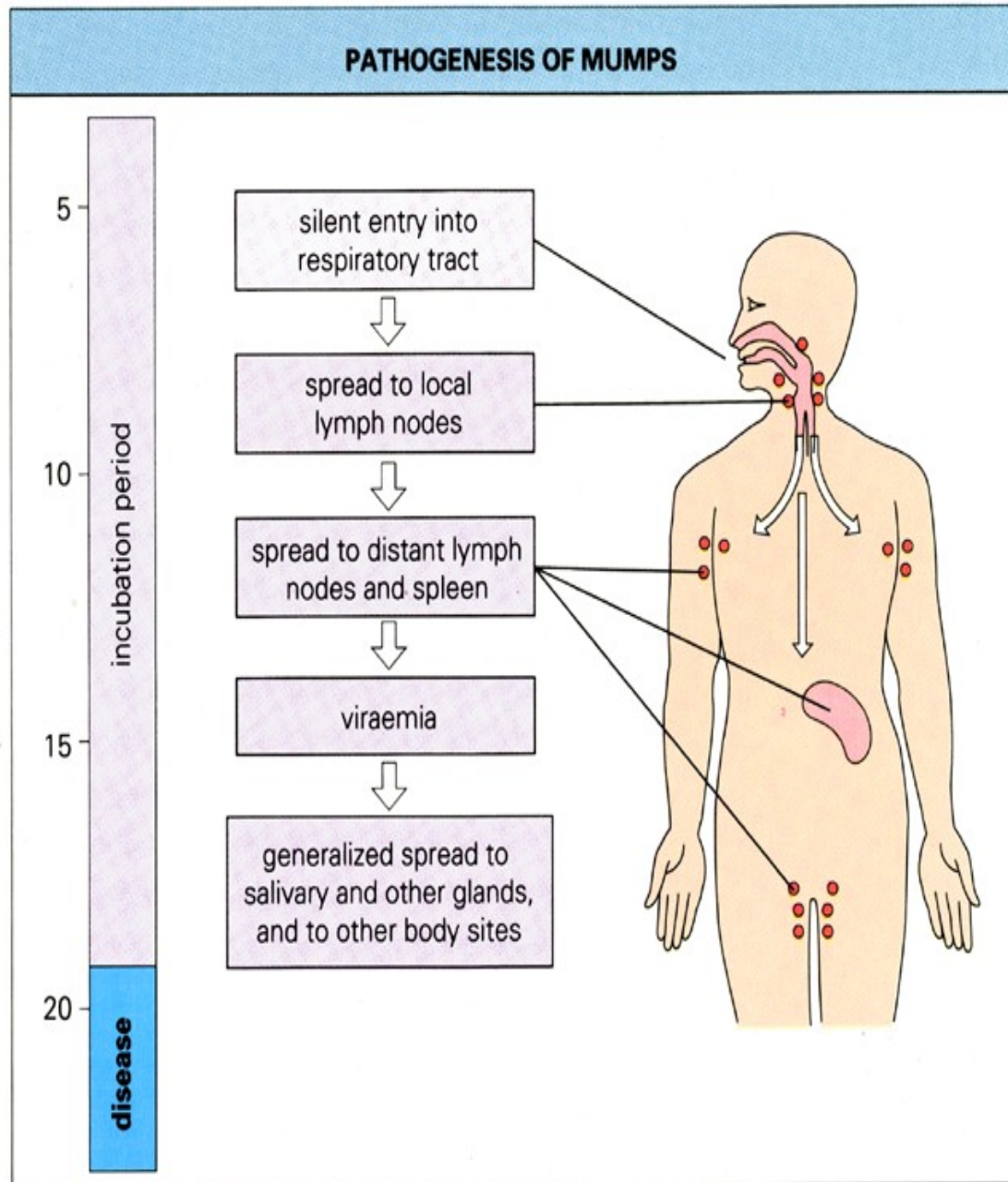
- Mammals & birds.
- strictly confined to the **mucosal cells** of the respiratory tract
- Infected respiratory epithelial cells become vacuolated, show damaged cilia, and may form syncytia
- Cell damage triggers the production of inflammatory mediators
 - nasal secretion and cause local inflammation and swelling
- In humans, they cause:
 - Respiratory infections (common), including Severe Acute Respiratory Syndrome (SARS)
 - Enteric infections (occasional - mostly in infants <12 months)
 - Neurological syndromes (rare)

Mumps, Measles, Rubella, and Other Childhood Exanthems

- They are genetically unrelated
- they share several common epidemiologic characteristics:
 - distribution is worldwide  nonimmune individuals
 - humans sole reservoir
 - person-to-person spread primarily by aerosols

MUMPS

- a paramyxovirus, with only one antigenic type
- contains single-stranded, **- RNA** and enveloped.
- There are two **glycoproteins** on the surface of the envelope
 - one mediates neuraminidase and hemagglutination activity
 - the other is responsible for lipid membrane fusion to the host cell
- “Acute contagious disease characterized by nonsuppurative enlargement of one or both **paratoid glands**”
 - Viremia and Viruria are common
 - Salivary gland swelling
 - Testes and ovaries after puberty



DIAGNOSIS

30% INFECTIONS SUB-CLINICAL

ISOLATION- saliva. CSF, urine
Monkey kidney cells

Serology – antibodies paired sera
ELISA. HI common

Measles

- Acute, highly infectious(**contagious**) disease characterized by **maculopapular rash**, fever and respiratory symptoms
- Paramyxoviridae, Genus- *Morbillivirus*
 - Enveloped, **ss** RNA virus
 - Glycoproteins H and F
 - F protein **highly conserved**
 - Has hemagglutinin but lacks neuraminidase activity
 - The receptor for **measles virus is CD46** (membrane cofactor protein), a regulator of complement activation
- Form **intranuclear inclusions**-other Paramyxoviruses do not
- Complications bacterial super-infection

Diagnosis

- Typical measles- clinical picture-diagnostic
- Atypical laboratory diagnosis
 - IgG and IgM antibodies-diagnostic
 - Fourfold increase titer acute and convalescent ELISA, Nt, HI
 - Demonstration of specific IgM antibodies
 - Isolation of virus-nasopharyngeal swabs/blood fever period best
 - Monkey/human kidney cells/ HEp-2- slow growth
 - CPE- giant cells with i/c and i/n inclusions

Rubella

- + sense RNA, enveloped virus, member of the **Togaviridae** family → 2 genera: alphaviruses and rubiviruses
- Among the most simple enveloped animal viruses with **Icosahedral symmetry**
- Rubella virus sole member of genus Rubiviruses
- Spread by respiratory droplets.
- In the prevaccination era, 80% of women were already infected by childbearing age.
- Limited host range-only humans
- Growth very slow in vertebrate cells too with minimum CPE
- **Rubella Infection**
 - **Postnatal rubella, German or three day measles**
 - **Congenital Rubella,**

Congenital Rubella results in:

- Normal independent lives or
- Baby low body weight, thrombopenia, hepatomegaly or
- Congenital abnormalities
- Cataracts, microphthalmia, heart defects, deafness. Mental retardation etc
- Time of fetal infection very important

Once fetus infected-The only effective way [terminate the pregnancy](#)

Prevention

Antenatal screening

- All pregnant women attending antenatal clinics are tested for immune status against rubella.
- Non-immune women are offered rubella vaccination in the immediate post partum period.
- Vaccine schedule 12-15 month age 1st dose of **MMR**

Diagnosis of acute infection

- Rising titres of antibody (mainly IgG) - HAI, EIA
- Presence of rubella-specific IgM – EIA